

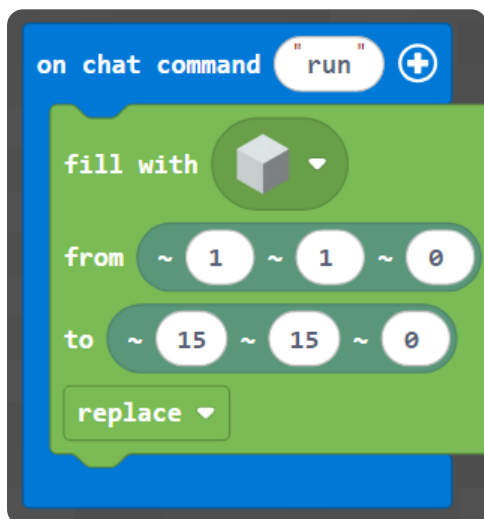
M003 Week 4 — English Worksheet (Intermediate)

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Intermediate · **Time:** about 40 minutes

This week you copy a **tree** onto your wall. Like last week, every block sits at a spot named by **two numbers** — (x, y). x is how far **across**, y is how far **up**. Read the picture, find each block's (x, y), and place it.

First, build your canvas. Run this to make a blank 15 × 15 wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot** — not from the corner of the world. Stand on the red block *every* time you run, or the whole picture shifts. Move your feet, move your picture.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write a full sentence saying what you think you will see.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
place brown block at (8, 4)
```

Across or up? Which number stays the same? Write a full sentence. (This is the tree's trunk.)


```
place orange block at (2, 15)
place orange block at (3, 15)
```

Across or up? Which number stays the same? Write a full sentence.


```
place green block at (8, 15)
place green block at (7, 14)
place green block at (9, 14)
```

These green blocks sit high and spread out as they go down. Write a full sentence: is this a fruit, or the top of the tree?

2 · Spot the Bug 🐛

Each block of code below is broken. Read what it is **supposed** to do, rewrite it so it works, then explain in 1 or 2 sentences why it was wrong and why your fix works.

Bug A — This should make the **tree trunk**: four brown blocks going **up** at $x = 8$, from $(8, 1)$ to $(8, 4)$. Right now they all land in the **same spot**.

```
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

Hint: to move up, the **y** has to change each time.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The tree trunk should grow **up** at $x = 8$: $(8, 2)$, $(8, 3)$, $(8, 4)$. Right now it grows **across** instead.

```
place brown block at (2, 8)
place brown block at (3, 8)
place brown block at (4, 8)
```

Hint: check which slot the changing number is in — the x slot (across) or the y slot (up). The two numbers got swapped.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Build It 📷

Switch to your homework world and stand on your **home spot** (the red block). Copy the **tree** below onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**, the numbers down the **side** are **y**.

🌳 Tree

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15						■	■	■	■	■					
14					■	■	■	■	■	■	■				
13				■	■	■	■	■	■	■	■	■			
12			■	■	■	■	■	■	■	■	■	■	■		
11			■	■	■	■	■	■	■	■	■	■	■		
10		■	■	■	■	■	■	■	■	■	■	■	■	■	
9		■	■	■	■	■	■	■	■	■	■	■	■	■	
8		■	■	■	■	■	■	■	■	■	■	■	■	■	
7			■	■	■	■	■	■	■	■	■	■	■		
6				■	■	■	■	■	■	■	■	■			
5						■	■	■	■	■					
4							■	■	■						
3							■	■	■						
2							■	■	■						
1							■	■	■						

When your tree is finished, send a photo. Then explain what you did with these sentence starters — write 4 sentences.

First, I built the trunk by placing brown blocks at ...

The lowest block in my picture is at (x = ..., y = ...).

To go across a row I changed the ... number; to go up I changed the ... number.

The trickiest part to read off the grid was ...

First, I built the trunk by placing brown blocks at ...

The lowest block in my picture is at (x = ..., y = ...).

To go across a row I changed the ... number; to go up I changed the ... number.

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The trickiest part to read off the grid was ...

4 · Explain It 🎥

Take a video on a phone while you show your tree on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **x, y, coordinate, across, up, home spot**.

1. Show your home spot (the red block) and say why you stand on it every time.
2. Show your finished tree from the front.
3. Point at one block and read its coordinate out loud: "this one is at x ..., y ...".
4. Show one block you placed in the wrong spot at first, and how you fixed it.

Write about 4 sentences saying what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.